

TRISTAN ALDERSON

Work Authorization: Dual US & Canadian Citizen

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EDUCATION

Bachelor of Applied Science in Electrical and Computer Engineering

Queen's University • *Dean's Scholar*

Kingston, ON

05/2026

WORK EXPERIENCE

Avionics Lead

05/2025 – Present

Queen's Rocket Engineering Team (QRET)

Kingston, ON, Canada

- **Lead 10 engineers** building the distributed avionics network for a hybrid rocket. Wrote a fixed-format **CAN** protocol to sync time and forward telemetry across **STM32** and **ESP32** nodes.
- Designed **STM32** and **ESP32** control boards with hardware fail-safes for solenoid valves and switched power logic to cut idle draw on pressure transducers.
- Engineered a 24V power module that ORs ground supply and a 18650-based boost converter; sized bucks for CCM and added PMOSFET reverse polarity protection with a zener shunt.
- Reworked a custom handheld **ground station** receiver with live telemetry decoding and a 330mF supercapacitor to hold GPS locks through brief power losses.
- Managed **PCB design cycle** from **KiCad** schematics through JLCPCB assembly, manual rework, and bench bring-up.

Embedded Software Engineer Intern

05/2024 – 08/2025

Evertz Microsystems

Burlington, ON, Canada

- Designed distributed backend control software in **C++** on **Zynq UltraScale+ MPSoCs**, coordinating status and telemetry across multiple ARM Linux nodes.
- Debugged 10G/100G **SFP/QSFP** data paths, **multicast** forwarding, and switch behavior using Marvell vendor tooling, **GDB**, logs, and board-level traces.

AI Engineer & Avionics Developer

10/2023 – 08/2024

Queen's Rocket Engineering Team (QRET)

Kingston, ON, Canada

- Secured **2nd Place at Launch Canada** by implementing **YOLOv8** computer vision and **PID** control, simulated using **Matlab**, for an automated rocket tracking system.
- Served on a 10-person launch team in New Mexico, managing **18650 battery management systems** and critical altimeter/GPS configurations.
- Designed and flew a custom **SAM-M10Q GPS** daughter board; isolated it from the main comms stack to cut RF noise and enlarged the ground plane to hold locks during high-G flight.

Programmer and Assembler

05/2023 – 09/2023

Radius Security

Richmond, BC, Canada

- Manufactured remote-guarding surveillance units, earning an internal achievement for wire management quality.
- Contributed to **Lean** process improvements that increased manufacturing space and overall **efficiency by 20%**.

PROJECTS

LiteLM: On-Device LLM Inference (Capstone Project)

08/2025 – 03/2026

Queen's University

Kingston, ON

- Earned **3rd Place for Top Computer Engineering Project** at the Queen's ECE Showcase for running GPT-2 inference entirely on the **AMD Xilinx Kria KV260** edge hardware.
- Architected **AXI4 DMA** data paths: MM2S streams token embeddings from DDR to the accelerator while S2MM writes output tokens back.
- Wrote custom **Linux kernel drivers** and device trees to load the hardware overlay and manage transfers between the ARM Cortex-A53 and FPGA fabric.

SKILLS

Hardware & EDA: KiCad, LTspice, Vivado, Vitis HLS, oscilloscopes, logic analyzers

Embedded Platforms: STM32, ESP32, Zynq UltraScale+ MPSoCs, Kria KV260, x86 COM Express, ARM Linux

Languages: C/C++, Python, Bash, Matlab, Verilog/SystemVerilog

Systems & Tools: Embedded Linux, RTOS, Buildroot, U-Boot, device trees, GDB, Git, Prometheus, Grafana

Protocols & Bus: AXI4 (Lite, Full, Stream), DMA, CAN, I2C, LoRa, NTP/PTP, 1G/10G/100G SFP/QSFP, ACL/TCAM